4 **(a)** On the axes in the **Printed Answer Booklet**, sketch the graph of $y = \sin 2\theta$ for $0 \leqslant \theta \leqslant 2\pi$. **[2]**

(b) Solve the equation $\sin 2\theta = -\frac{1}{2}$ for $0 \leqslant \theta \leqslant 2\pi$. **[3]**

